Machine Learning for Crop Disease Detection

Master's Thesis

Author: Carol Williams
Email: student3@university.edu

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Abstract

This thesis investigates the application of deep learning techniques for automated detection and classification of plant diseases from leaf images. Agricultural losses due to plant diseases significantly impact food security and farmer livelihoods worldwide.

The research develops and evaluates multiple convolutional neural network architectures for disease classification, achieving 94.2% accuracy on a dataset of 54,000 images spanning 38 disease categories across 14 crop species.

The key contributions include:
- A comprehensive survey of existing plant disease detection methods
- Novel data augmentation techniques for agricultural imagery
- Comparative analysis of CNN architectures for this domain
- A mobile application prototype for field deployment
- Recommendations for practical implementation in farming contexts

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Chapter 1
Introduction

Plant diseases cause estimated annual losses of $220 billion globally. Early detection and treatment can significantly reduce these losses. Traditional disease identification relies on expert knowledge, which is often unavailable in rural farming communities. This research explores automated detection using machine learning.

This section contains additional technical details and supporting information relevant to the chapter topic. The content demonstrates the depth of research and analysis conducted throughout this study.

Further considerations include implementation specifics, potential limitations, and suggestions for future improvements. These elements contribute to the comprehensive nature of this academic work.

Chapter 2
Literature Review

Previous work in plant disease detection has utilized various approaches including traditional machine learning with handcrafted features, transfer learning from ImageNet-pretrained models, and custom CNN architectures. This chapter reviews 50 relevant publications from 2015-2023.

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Chapter 3
Methodology

The research methodology includes dataset preparation, model architecture design, training procedures, and evaluation metrics. The PlantVillage dataset was augmented with images collected from local farms to improve model generalization.

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Chapter 4
Dataset Analysis

The combined dataset contains 54,306 images across 14 crop species and 38 disease categories. Class imbalance was addressed through oversampling and weighted loss functions. Data augmentation included rotation, flipping, color jittering, and synthetic occlusions.

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Chapter 5
Model Architecture

Several architectures were evaluated: VGG16, ResNet50, InceptionV3, and a custom lightweight model for mobile deployment. The best performing model uses EfficientNet-B4 as the backbone with custom classification layers.

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Chapter 6
Experiments and Results

Training was performed on NVIDIA RTX 3090 GPUs using PyTorch. The best model achieved 94.2% accuracy, with per-class F1 scores ranging from 0.87 to 0.99. Confusion matrix analysis revealed common misclassifications between visually similar diseases.

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Chapter 7
Mobile Application

A prototype Android application was developed using TensorFlow Lite for on-device inference. The quantized model achieves 91.5% accuracy with inference time under 200ms on mid-range smartphones.

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Chapter 8
Discussion

The results demonstrate the feasibility of automated plant disease detection. Limitations include performance degradation under poor lighting conditions and with severely damaged leaves. Field trials showed 89% user satisfaction among farmers.

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Chapter 9
Conclusion and Future Work

This thesis contributes a practical solution for plant disease detection. Future work includes expanding the dataset to cover more crop species, implementing disease severity estimation, and integrating with agricultural advisory systems.

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References

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